

Name:		Index Number:		Class:	
-------	--	---------------	--	--------	--



DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

H1 BIOLOGY

8875/02

Paper 2 Structured and Free-response (Booklet I)

16 Sep 2011

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A:

Consists of 6 Structured Questions separated into Booklets I and II.

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section B:

Consists of 2 Essay Questions found in Booklet II.

Answer any **one** question.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

Do not open the booklets until you are told to do so.

BOOKLET I		
FOR EXAMINER'S USE		
Section A1	1	/5
	2	/5
	3	/5

This document consists of **5** printed pages and **1** blank page.

Booklet I

Section A1: Structured Questions (15 marks)

Answer **all** questions in this section.

Question 1

For
Examiner's
use

Colour blindness is controlled by a gene on the X chromosome. The allele for colour blindness, X^b , is recessive to the allele for normal colour vision, X^B .

- (a) Explain why colour blindness is more common in men than in women. [2]

- (b) The gene controlling the presence of a white streak in the hair is not sex linked, with the allele for the presence of a white streak, **H**, being dominant to the allele for the absence of a white streak, **h**. **Fig. 1.1** shows a family tree in which some of the individuals have a white streak present in the hair.

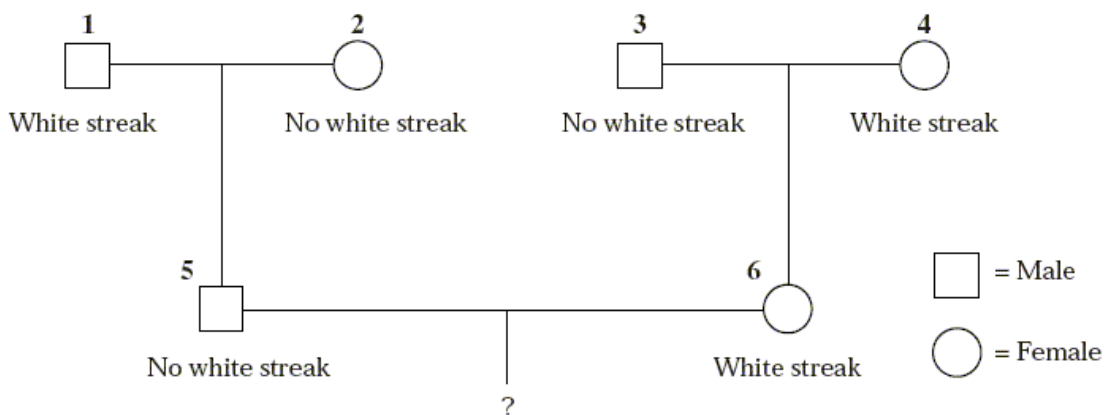


Fig. 1.1

- (i) What are the genotypes of individuals 5 and 6? [1]

Individual 5:

Individual 6:

- (ii) Give the possible genotypes of the gametes produced by individuals 5 and 6? [1]

Individual 5:

Individual 6:

- (iii) What is the probability that the first child of individuals 5 and 6 will be a boy with a white streak in his hair? Show your working. [1]

Total: [5]

Question 2

The bone protein, osteocalcin, has been extracted from fossil Neanderthal skulls found in Shanidar Cave, Iraq. The skulls were estimated to be approximately 75 000 years old.

Using mass spectrometry, the amino acids of this Neanderthal osteocalcin have been sequenced. This sequence has been compared with that in osteocalcin from *Homo sapiens* and other modern primates.

The sequences of the first 20 amino acids in the primary structure of osteocalcin from this study are shown below. Amino acids are represented by capital letters.

	1	10	20
Modern human	Y	L	Y
Neanderthal	Y	L	Y
Chimpanzee	Y	L	Y
Orang-utan	Y	L	Y
Gorilla	Y	L	Y
Monkey	Y	L	Y

- (a) Suggest what the data indicate about the relationships between Neanderthals and modern primates. Give reasons for your answer. [3]

- (b) Explain how material from the Neanderthal skulls could be used to estimate their age. [2]

Total: [5]

Question 3

- (a) Chymosin is an enzyme used to coagulate milk in cheese-making. It was obtained from the stomachs of calves. Now it can be made using gene technology.

The scientists who first used gene technology to produce this enzyme inserted the chymosin gene into plasmids. The plasmids were then mixed with yeast cells.

Some of the yeast cells took up the plasmids and some did not. The yeast cells which took up the plasmids produced chymosin.

- (i) What is a marker gene? [1]

- (ii) A marker gene could be used to obtain a sample of yeast cells which all contain the chymosin gene. Explain how this is possible. [2]

- (iii) Antibiotic resistance genes are often used as marker genes. Explain why some people are concerned about this use of antibiotic resistance genes. [1]

- (iv) Suggest **ONE** advantage of making cheese with chymosin from yeast, rather than from calves. [1]

Total: [5]

END OF BOOKLET I

BLANK PAGE

PLEASE DO NOT WRITE ON THIS PAGE

Name:		Index Number:		Class:	
-------	--	---------------	--	--------	--



DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 BIOLOGY

8875/02

Paper 2 Structured and Free-response (Booklet II)

16 Sep 2011

2 hours

Additional Materials: Writing paper

BOOKLET II		
FOR EXAMINER'S USE		
Section A2	4	/7
	5	/11
	6	/7
Section B	7 / 8*	/20

* Please circle the appropriate number.

This document consists of **9** printed pages and **1** blank page.

Section A2: Structured Questions (25 marks)

Answer **all** questions in this section.

For
Examiner's
use

Question 4

Fig. 4.1 below shows how the amount of DNA in a cell of *Chrysanthemum makinoi* varies during periods **P** to **T** in the cell cycle.

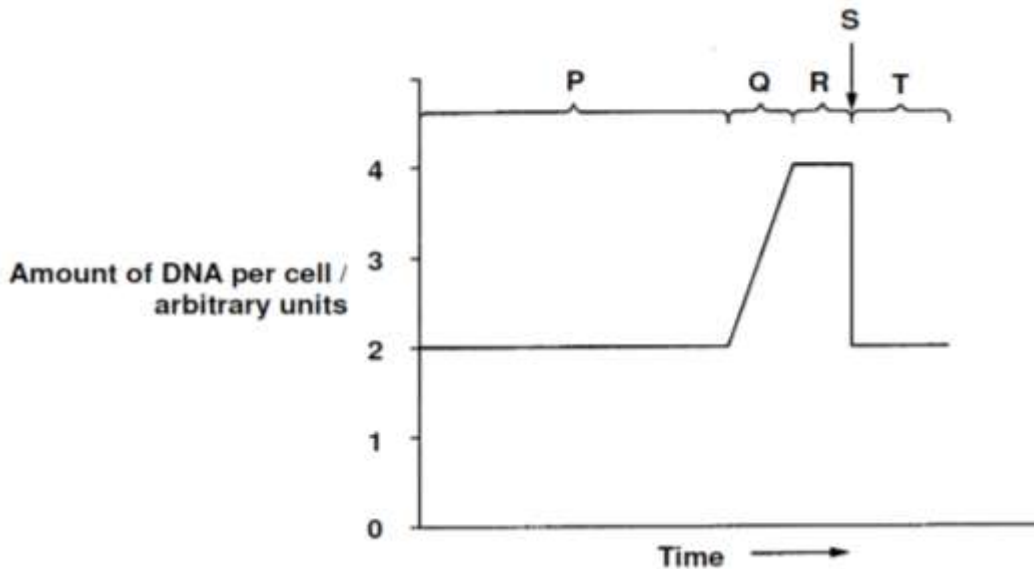


Fig. 4.1

- (a) Radioactive thymine was supplied to the cells of some growing *Chrysanthemum* tissue.

With reference to **Fig. 4.1**, during which period (**P** to **T**) of the cell cycle does radioactivity of the nuclei first increase? Explain your answer. [3]

- (b) At anaphase of mitosis, sister chromatids separate to become individual chromosomes.

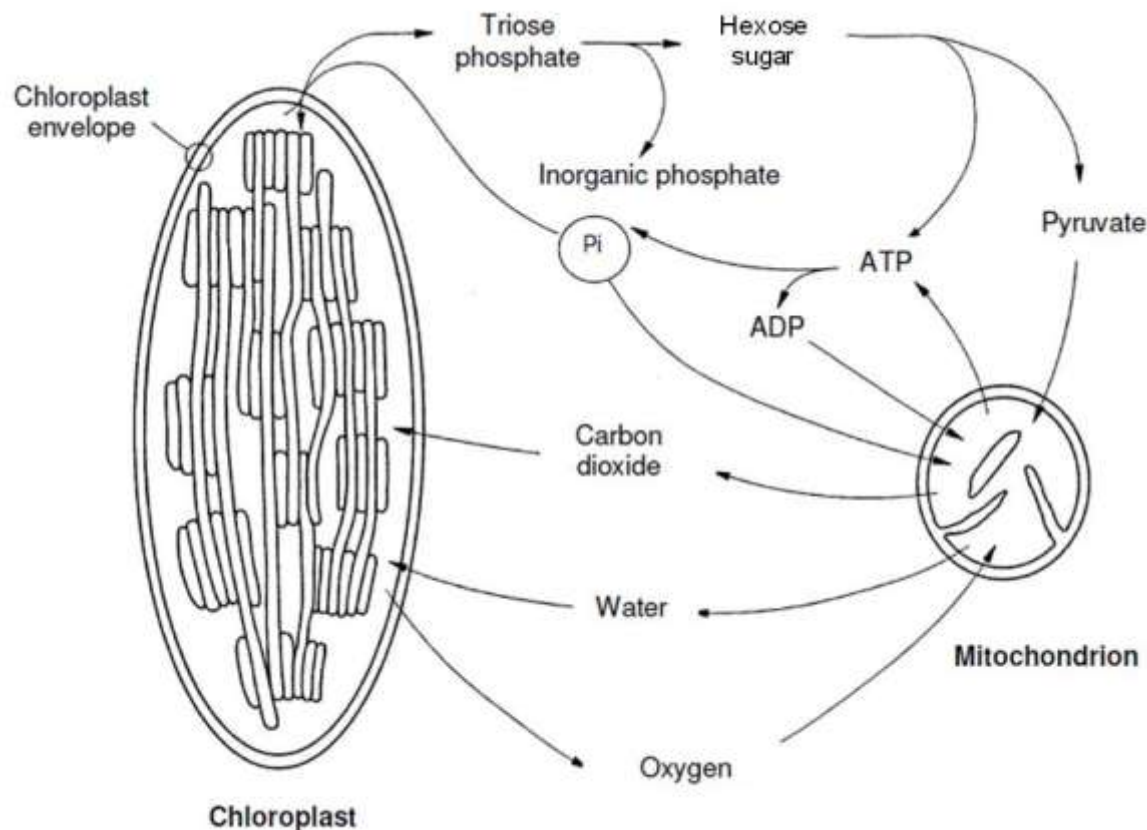
- (i) Distinguish between **chromatids** and **chromosomes**. [2]

- (ii) Identify the period (**P** to **T**) in **Fig. 4.1** during which anaphase occurs. Briefly explain your answer. [2]

Total: [7]

Question 5

Fig. 5.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.

**Fig. 5.1**

A leafy shoot is sealed inside an air-tight transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Fig. 5.1**, outline how the following will link the metabolic pathways between the chloroplast and mitochondrion of this cell:

(i) triose phosphate and hexose sugar; [2]

- (ii) gaseous exchange. [2]

- (b) With reference to **Fig. 5.1**,

- (i) explain how it is possible for the oxygen concentration to remain constant. [2]

- (ii) explain why there is no build up in the concentration of phosphate ions inside the mitochondria as a result of the inward passage of phosphate ions. [2]

- (c) An aqueous solution contains small membranous vesicles which may be derived either from a thylakoid or a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 5.2 below shows the various experimental systems that are prepared in an attempt to synthesize ATP.

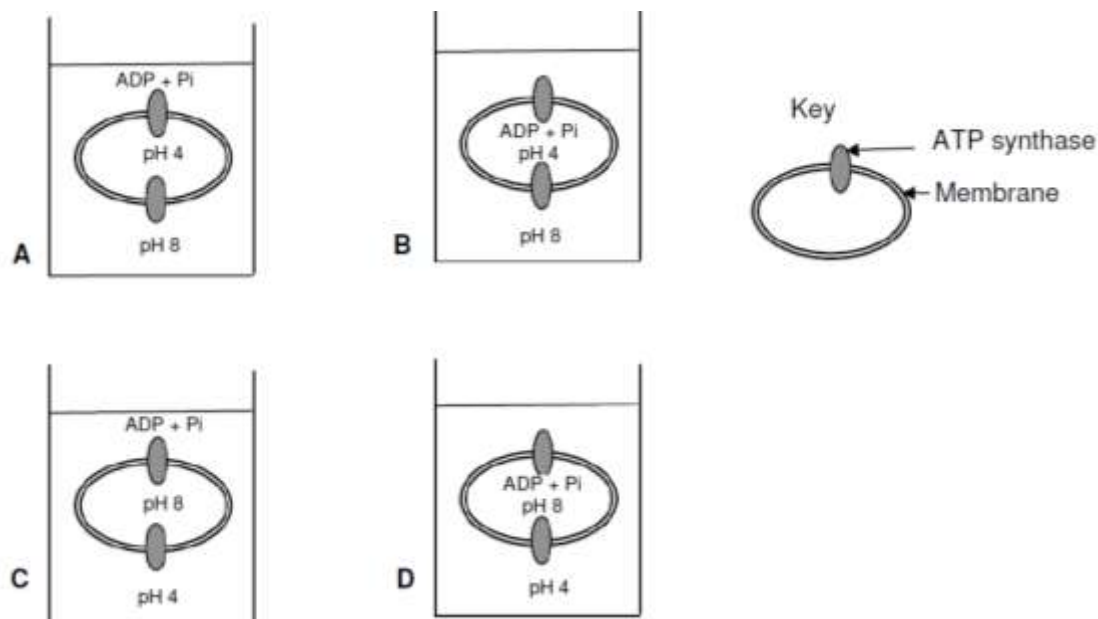


Fig. 5.2

Which experimental system (A to D) shown in **Fig. 5.2** would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer. [3]

Total: [11]

BLANK PAGE

PLEASE DO NOT WRITE ON THIS PAGE

Question 6

The bacterium *Bacillus thuringiensis* (Bt) makes the Bt-toxin which is toxic to *Lepidopteran* caterpillars and has no effect on most other groups of insects, invertebrates and vertebrates (including Man). The dried bacteria have been used as a biological control measure, sprayed in suspension onto crop plants. However, it becomes ineffective after a few days.

The ideal solution would be to have the crop plants make their Bt-toxin. Since Bt-toxin is a protein, this is quite feasible using genetic engineering techniques.

Fig. 6.1 below shows how tomato plants are genetically engineered to produce their own toxin.

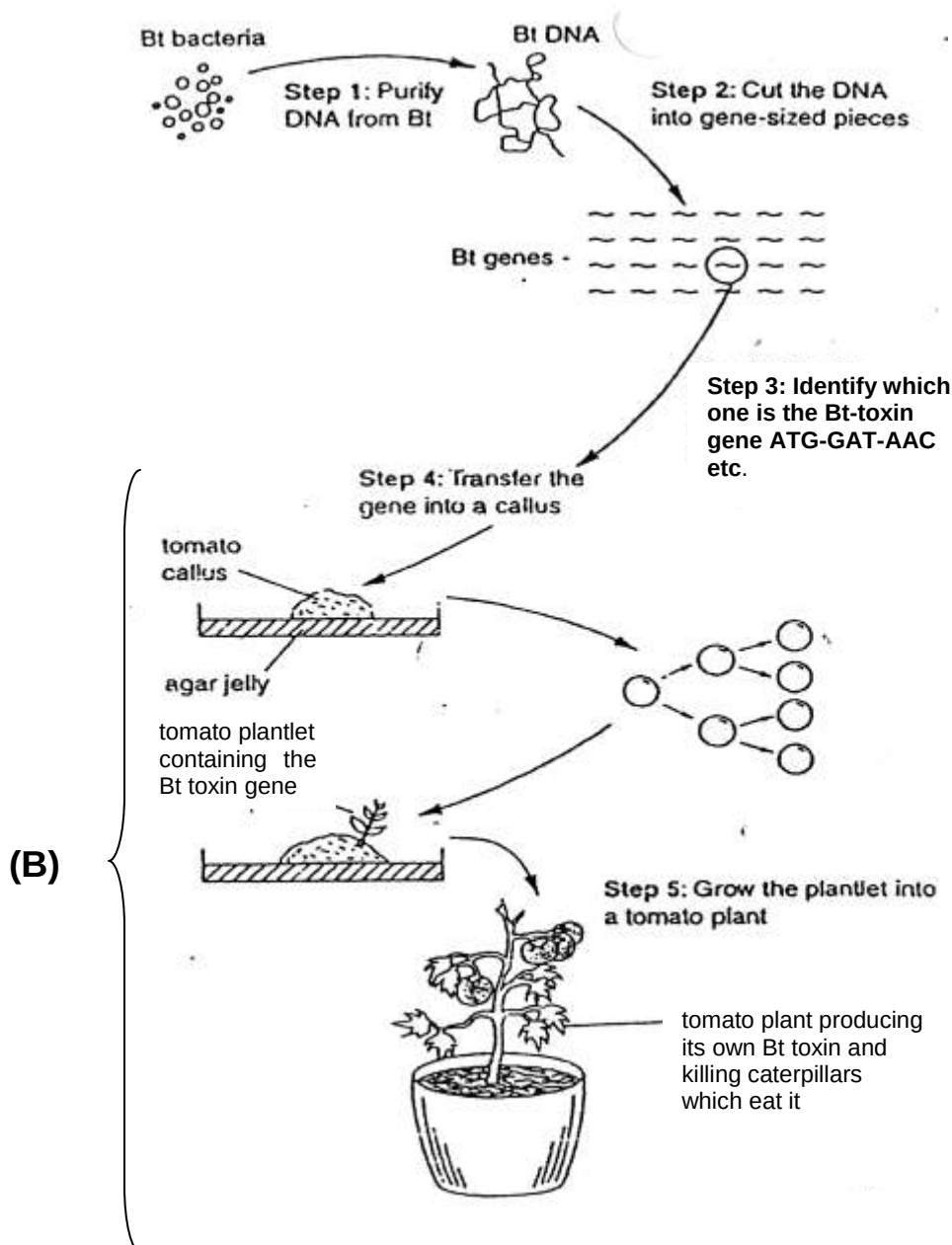


Fig. 6.1

- (a) Suggest why it was considered a better option to engineer crop plants to make the toxin, rather than to spray the plants with dried bacteria. [1]

- (b) With reference to the portion marked **(B)** in **Fig. 6.1**, state and explain fully whether the cells in the callus are totipotent or specialised. [4]

- (c) Explain **two** possible hazards of genetic engineering of Bt crop plants. [2]

Total: [7]

Section B: Essay Questions (20 marks)

Answer only **one** question.

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 7

- (a)** The phospholipid bilayer forms an essential part of living cells but can also serve as a hindrance to certain cellular processes. Highlight two distinct examples and explain how the cell copes with this problem. [6]
- (b)** Explain how the fatty acids of phospholipids differ in membranes of tissue exposed to low temperatures and those found at higher temperatures. [5]
- (c)** Describe the experimental evidence for semi-conservative replication. [9]

OR

Question 8

- (a)** Describe, with named examples, how the **three** forms of natural selection bring about evolution. [10]
- (b)** Explain how gel electrophoresis is used to separate DNA. [5]
- (c)** Describe the disadvantages of cloning eukaryotic genes in bacterial cells. [5]

[Total: 20]

END OF PAPER